

ECON 100B: Macroeconomics

Class 3: The Long Run (Jones Chapter 3)

Ryan D. Edwards
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Agenda

- Announcements
- Exam logistics shown in its instructions on page 1
- A quick look back at where we've been (Chapters 1 & 2)
- “The Long Run” (Chapter 3) a.k.a. An Overview of Long-Run Economic Growth

Announcements

- Please complete the Plagiarism+ Quiz on bCourses by next **Friday February 6**
- PS 1 is now available and will be due on Gradescope on **Friday February 13**

A great question

- “What if we never cover material that appears in the slides? Is that still fair game for the exams?”
- If the material is in the slides, chances are good that you should study it and ask about it in section or office hours.
- It's material that I think will help you learn the core concepts
- But it's never the core concepts themselves. We will cover those during class, for sure. Core concepts are probably what you should write down on your cheat sheet

Looking ahead to the midterm exam: The first page of each exam shows the rules.

1. Stop when time is called, or face stacking 10 point penalties
2. Give your completed exam to a proctor or into the receiving box
3. Open-book, **one page front & back of notes.** Calculator. Ruler.
4. No Internet, no comms, no screens
5. Sign honor code
6. Write legibly
7. Restroom breaks: check with a proctor. Give them your phone
8. If you see something, say something

The first page is also shown inside our folder on Drive

Outline of Macroeconomics

- Chapter 2 shows how we measure income, $Y = C + I + G + NX$
- **Chapters 3–6 discuss long-run growth**
- Chapters 7–8 talk about inflation and unemployment
- Chapters 9–14 discuss short-run fluctuations, fiscal policy, monetary policy, the Phillips Curve, the Fed
- Chapters 16 & 17 focus on Consumption and Investment
- Chapter 18 looks at Government
- Chapters 19 & 20 address international trade and finance, exchange rates

Learning objectives in Chapter 3

- Economic growth is a fairly recent phenomenon. If all of human history took place in a year, growth only happens on Dec 31
- But once it starts, economic growth is **exponential**, not linear
- We mean growth in real GDP per person, which has been about 2% per year in the U.S. since about 1870
- Growth rate algebra and rules are key in ECON 100B
 - If y is growing at rate g , then $y_t = y_0 (1 + g)^t$ or $y_t = y_0 e^{gt}$
 - This is the same as $g = \frac{y_1 - y_0}{y_0}$ when $t = 1$
 - The **growth rate rules** operate the same as logarithm rules. For example, the growth rate of a product equals the sum of the growth rates
- Constant growth is convex on a linear scale but straight on a **ratio scale**

Teaser preview of a key part of Chapter 3

- If GDP grew by an annualized 3.2% rate,
- And if I told you that population is growing at a 0.7% annual rate,
- Then what is the growth rate in the ratio of GDP to population, a.k.a. GDP per capita?
- You might say: “I need a calculator.”
I would then reply: “No, grasshopper, you do not.”

ECON 100B so far: real GDP, real GDP per person, then compare the latter across geographic areas

- Using techniques described in the textbook, economists measure real GDP Y^R by removing price differences across time and space
- Real GDP Y^R and nominal GDP Y^N are related just like how quantity and total spending are related
- $P \times Q = \text{spending}$
- $P \times Y^R = Y^N$

Growth rates and ratios are friends

- Dataset from 1e shows in the U.S., real GDP per person Y^R/L was about \$30,000 in 2000, while it was about \$10,000 in 1950 when measured in 2000 dollars
- The average U.S resident in 2000 enjoyed a real standard of living that is **3 times higher** than that of the average resident in 1950
- For comparison: nominal GDP per capita in 1950 was \$2,000 in 1950 dollars, or **15 times lower** than the \$30,000 in 2000
- Real GDP per capita rose by a factor of 3, and if nominal GDP rose by a factor of 15, the price level must have risen by a factor of 5

A sketch of the U.S. around 1900

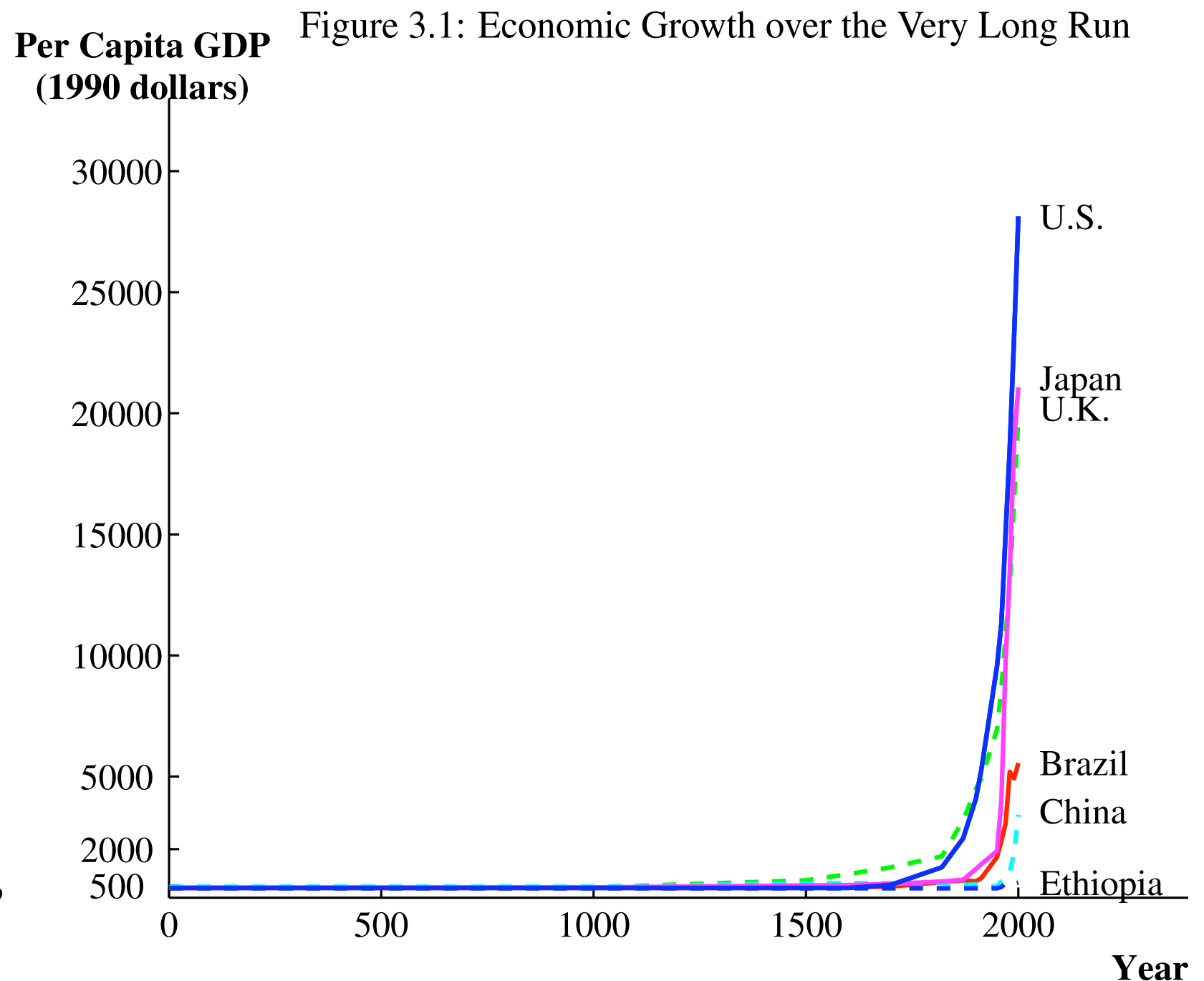
- Life expectancy at birth, the average **length of life** starting from birth, was about 50 [today: 77]
- One out of every 10 infants **died** before his or her first birthday [today: 7 out of every 1,000]
- 90 percent of households did not have **electricity**, refrigerator, telephone, or a car [today most do]
- Fewer than 10 percent of adults had graduated from **high school** [today: 85 percent]

A sobering comparison

	USA 1900	USA in 2000	Kenya in 2000
Life expectancy at birth	50	77	50
Infant mortality	0.1	0.007	0.06
Real GDP per capita (1990 \$)	\$4,100	\$28,000	\$1,000

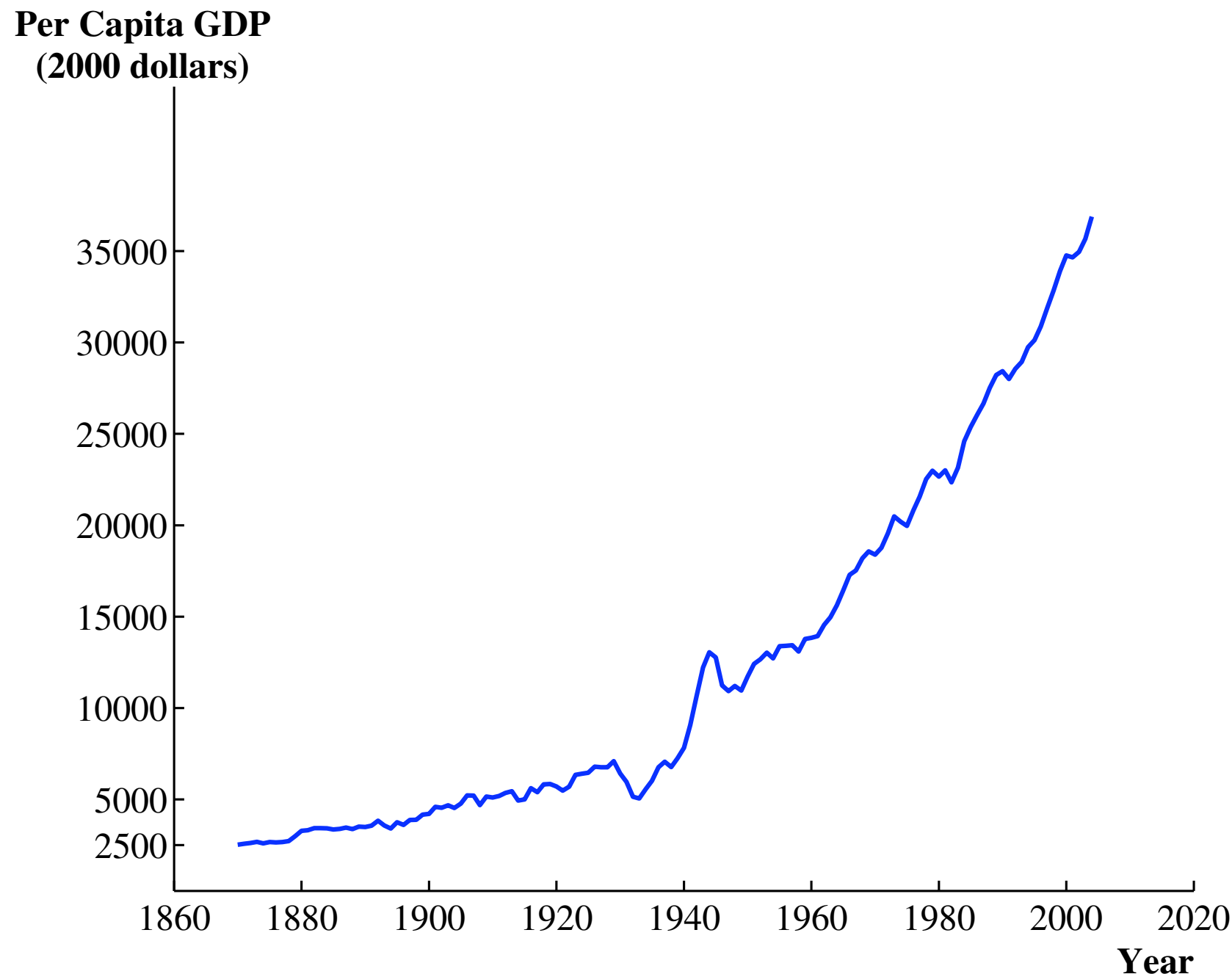
If growth has been uneven across geographic boundaries, what about across **time**?

- Growth is a relatively recent phenomenon, only in the past 2 or 3 centuries
- Growth arrived in different countries at different times
- Today, a “Great Divergence”



Note: Data from Angus Maddison, *The World Economy: Historical Statistics* (Paris: OECD Development Center, 2003).

Figure 3.2: Per Capita GDP in the United States



Note: Data from 1870 to 1928 from Maddison, cited in Figure 3.1. Data from 1929 to 2004 from the Bureau of Economic Analysis.

- Since 1870, GDP per person in the U.S. has risen 15-fold: from about \$2,500 to about \$37,000
- Another way to think about this is to compare **your** lifetime income with your parents'
- You'll be about **twice** as rich, because incomes have been doubling every 35 years, or roughly once every generation

The math of growth

- In Figure 3.2, GDP per capita is increasing by an *increasing* amount
- The increases are roughly proportional to the level at any point, say by some proportion g

$$y_{2005} - y_{2004} = \bar{g} \times y_{2004}$$

$$\frac{y_{2005} - y_{2004}}{y_{2004}} = \bar{g}.$$

- The left-hand side of the second equation, g , is the percentage change in GDP per capita, the growth rate. A little more math shows us:

$$y_{t+1} = y_t(1 + \bar{g}).$$

- With a constant growth rate, level increases are larger and larger over time
- Calculating levels using growth rates over a period of time is mathematically a little complicated
- Example: With 6.5 billion people in the world in 2006 and a constant annual growth rate of 2%, how large will world population be in 100 years, in 2106?
- Hint: It's **not**
$$6.5 \text{ billion} + 2\% \times 6.5 \text{ billion} \times 100 \text{ years}$$

(which equals 6.5 + 13 billion extra = 19.5 billion)

With a constant growth rate n , population after one year is $L_1 = L_0(1 + \bar{n})$.

After two years, it is $L_2 = L_1(1 + \bar{n})$.

Combining these, we find that

$$L_2 = L_0(1 + \bar{n})(1 + \bar{n}) = L_0(1 + \bar{n})^2.$$

So for some year t , $L_t = L_0(1 + \bar{n})^t$.

When $L_0 = 6.5$ billion and $n = 0.02$, $L_{100} = \mathbf{47.1 \text{ billion}}$
considerably larger than 19.5 billion!

We will also need to compute annual growth rates — but how?

- Rule of 70: if something is doubling every t years, you know that the growth rate is $70 \div t$

(With GDP per person, it's $70 \div 35$ years = 2%)

- Using the raw data: be a little careful!

$$y_t = y_0(1 + \bar{g})^t \rightarrow \boxed{\bar{g} = \left(\frac{y_t}{y_0}\right)^{1/t} - 1}$$

- You have to use a calculator or spreadsheet for this
- Note: this will give you 0.02 for 2% growth

Growth rate notation

These are all the same:

- The annual growth rate of x
- $(x_{t+1} - x_t) / x_t = x_{t+1}/x_t - 1$, or if the data are spaced further apart in time,

$$g_x = \left(\frac{x_t}{x_0} \right)^{1/t} - 1$$

- g_x or $g(x)$

All are numbers like 0.02 (which is 2%) per year

Properties of growth rates

1. Ratios become differences:

$$\text{If } z = x/y, \text{ then } g_z = g_x - g_y$$

2. Products become sums:

$$\text{If } z = x \times y, \text{ then } g_z = g_x + g_y$$

3. Powers become multiples:

$$\text{If } z = x^a, \text{ then } g_z = a \times g_x$$

(If this looks like logarithms to you, that's no accident!)

A key example that will turn up soon:

- Suppose we know that

$$Y_t = A_t K_t^{1/3} L_t^{2/3}$$

- What is the growth rate of Y_t in terms of the growth rates of A_t , K_t , and L_t ?
- The growth rate of a product is the sum of the growth rates

$$g(Y_t) = g(A_t) + g(K_t^{1/3}) + g(L_t^{2/3})$$

$$g(Y_t) = g(A_t) + g(K_t^{1/3}) + g(L_t^{2/3})$$

- The growth rate of a power is the power times the growth rate

$$g(Y_t) = g(A_t) + \frac{1}{3} \times g(K_t) + \frac{2}{3} \times g(L_t)$$

- We will later learn about this function; it tells us that growth in income (Y) comes from
 - Growth in productivity (A) plus
 - Growth in physical inputs (capital, K; and labor, L)

A real-world example

- Suppose you know the following:
- In 2010, you worked **$h = 20$** hours a week for **$k = 50$** weeks and earned **$w = \$10$** per hour
- So your total earnings **E** were equal to **$E = h k w$**
- With your earnings, you buy donuts priced at **$P = \$2$** per donut
- In 2011, times are tough!
- You can only get **$h' = 18$** hours per week for **$k' = 50$** weeks, and you didn't get a raise, so **$w' = \$10$**
- But times are also tough for Dunkin' Donuts, who have to cut donut prices to **$P' = \$1.80$** to stay competitive in a down market

Questions you can answer using analytical tools, *rather than a calculator*

	hours h	weeks k	wage w	price P
2010	20	50	\$10	\$2
2011	18	50	\$10	\$1.8

earnings **E = h k w**

- Q: Has your **real wage** (**w^r = w ÷ P**) risen or fallen?
 - Risen because **w** stayed the same while **P** fell. How much?
 - $g[w^r] = g[w] - g[P] = 0\% - (-10\%) = \underline{10\%}$ because $g[P] = -10\%$
- Q: Have your **real earnings** (**R = E ÷ P**) risen or fallen?
 - $g[E \div P] = g[E] - g[P] = g[h k w] - g[P] = g[h] + g[k] + g[w] - g[P] =$
 $= -10\% + 0\% + 0\% - (-10\%) = \underline{0\%}$

Your real wage rose by the same rate that your hours fell, so your real earnings are unchanged

Review: The strange and subtle mathematical appearance of growth rates

- If $x_t = (1.05)^t$
- What is the growth rate of x , also called g_x ?
- If you want to, you can compute this long-hand:
 - $x_0 = (1.05)^0 = 1$
 - $x_1 = (1.05)^1 = 1.05$
 - So $\frac{x_1 - x_0}{x_0} = \frac{1.05 - 1}{1} = 0.05$
- Or just memorize/recognize this: $x_t = (1 + g_x)^t$
- *But don't use the "growth rate rules" here!*

Plotting on a ratio scale (a.k.a. log scale)

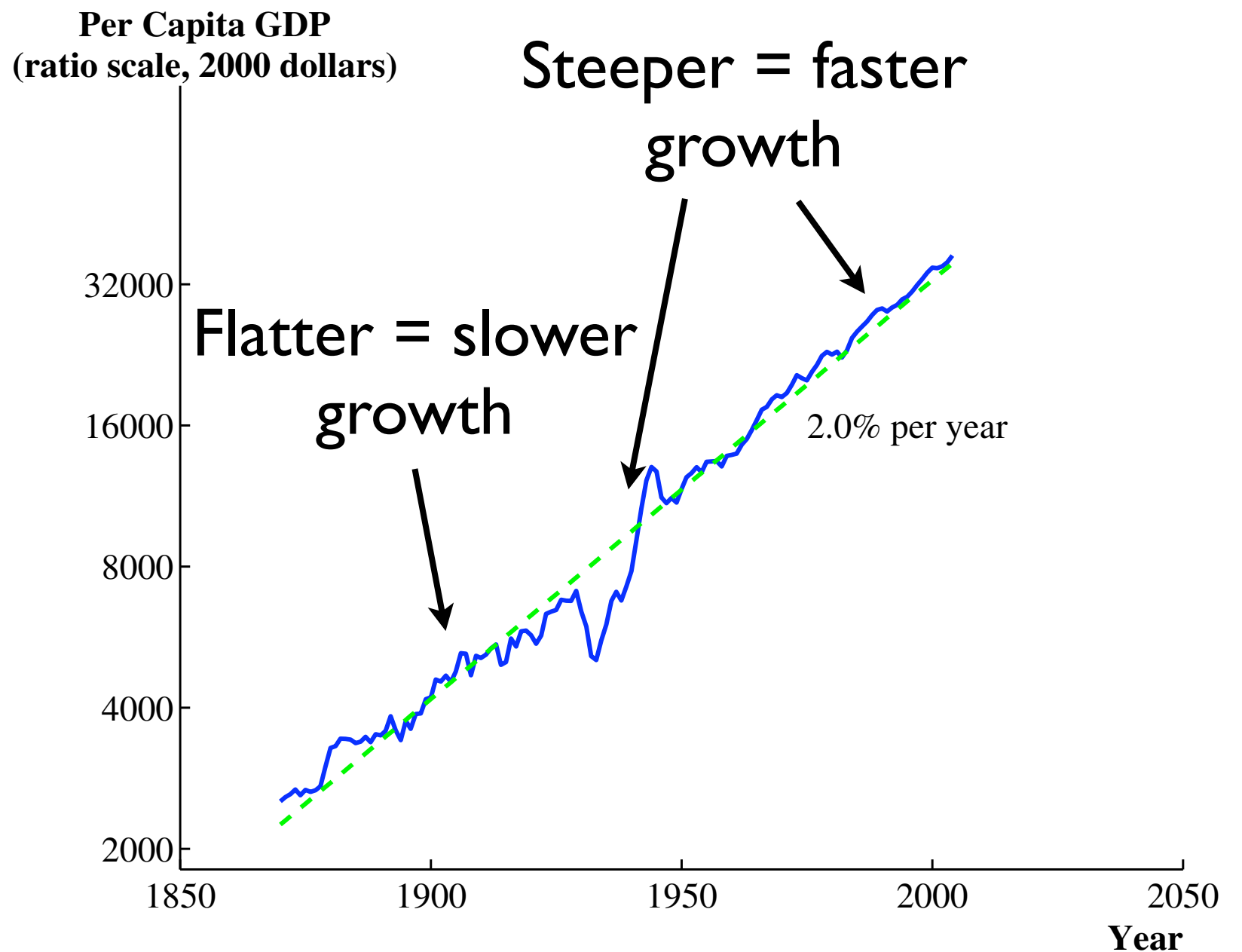
- If the y -axis is scaled in terms of ratios or multiples of an amount rather than its levels,
- Then a series that grows at a constant rate...
appears as a straight line on a ratio scale

On a ratio scale,
equal spacings
are constant
ratios (here, 2:1
or doubling)

GDP per person
has grown at a
fairly constant
rate of 2%

The slopes reveal
faster or slower
growth

Figure 3.5: Per Capita GDP in the United States: Ratio Scale



Note: This is the same data shown in Figure 3.2, but plotted using a *ratio scale*. Notice that the ratios of the equally-spaced labels on the vertical axis are all the same, in this case equal to 2. The dashed line exhibits constant growth at a rate of 2.0 percent per year.

Ratio scales allow us to see and tell stories about shifting growth rates much easier

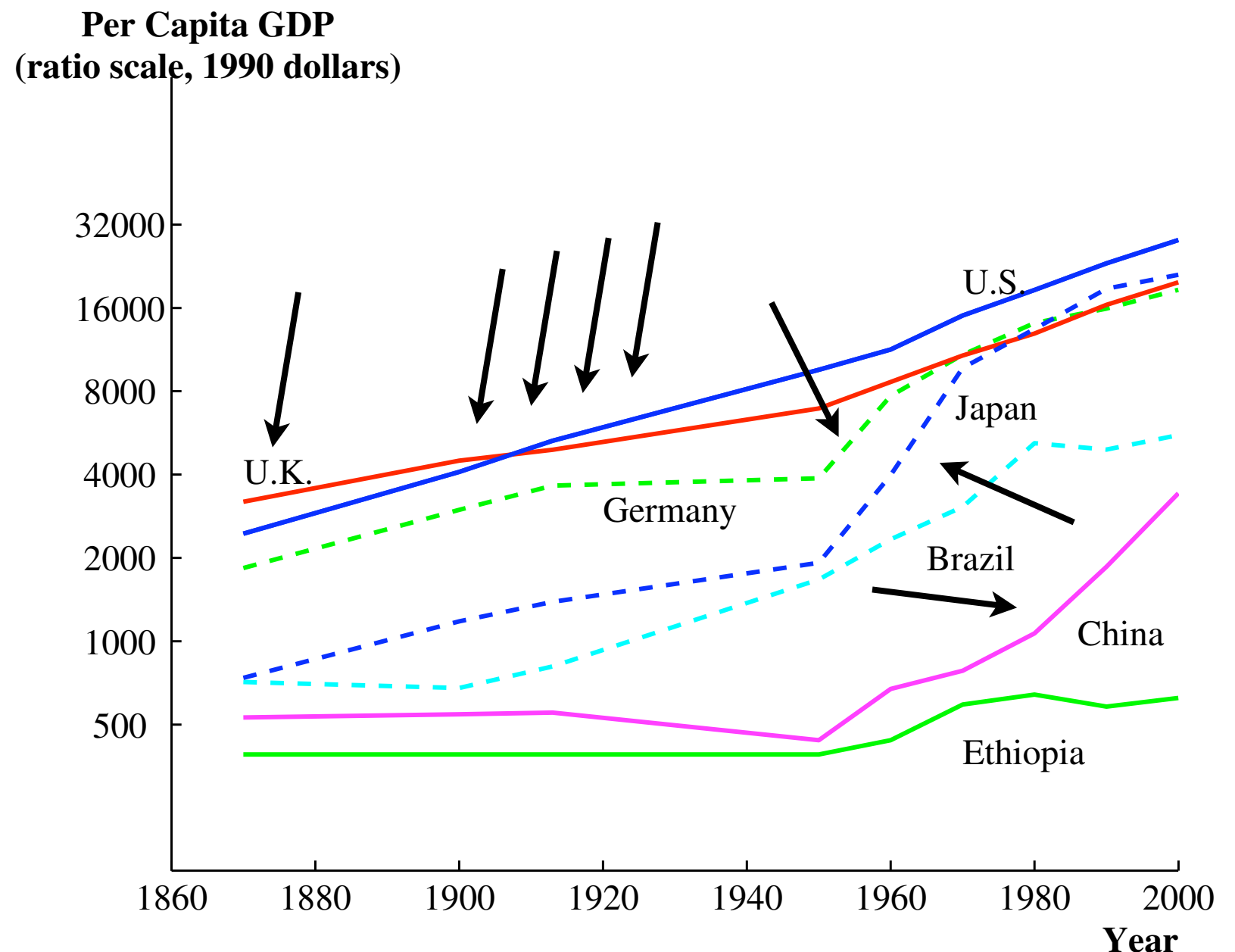
In 1870, the UK was the richest country

But the U.S. grew more rapidly!

Postwar Germany and Japan caught up

China is growing fast!

Figure 3.6: Per Capita GDP since 1870



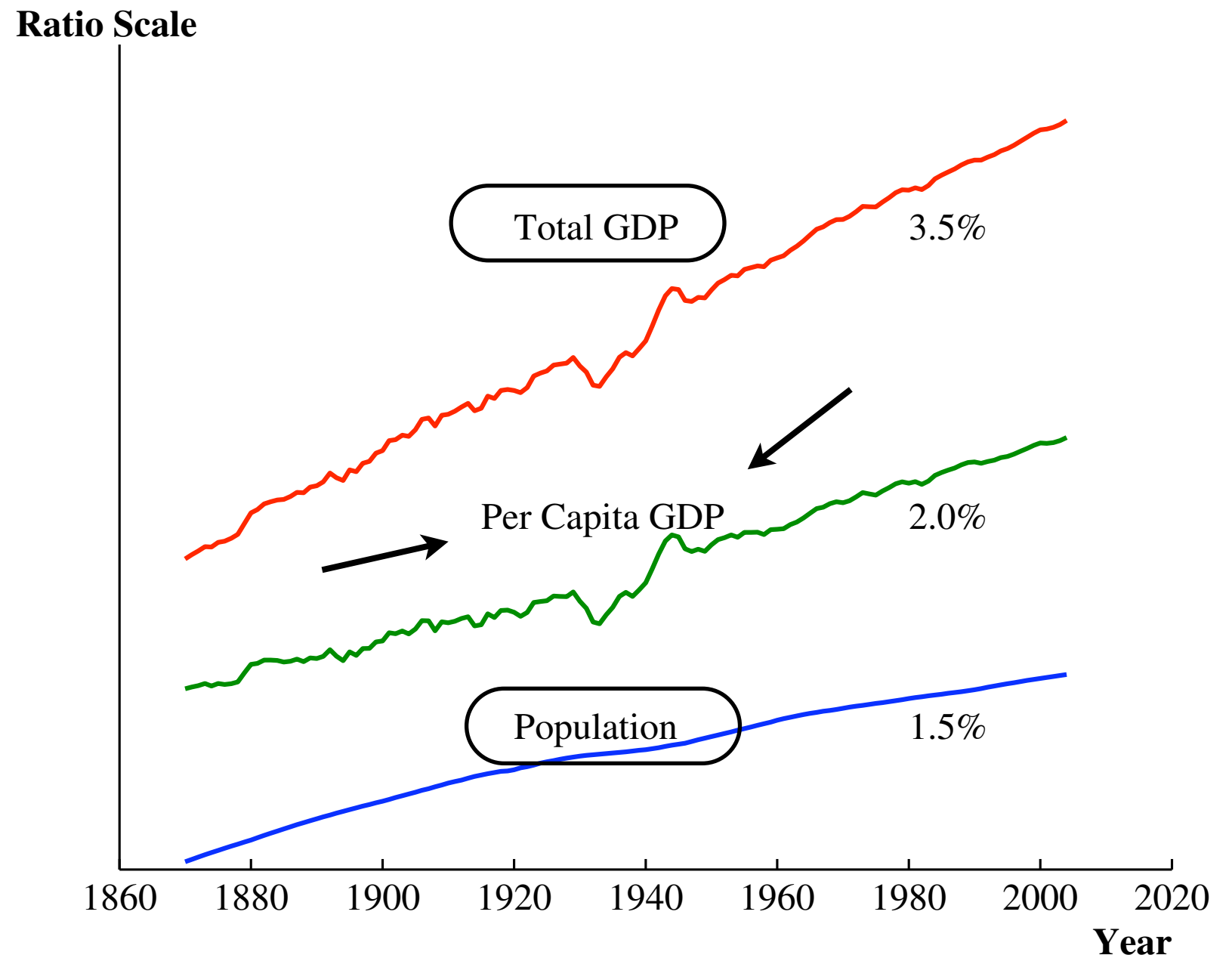
Note: Data from Angus Maddison, *The World Economy: Historical Statistics* (Paris: OECD Development Center, 2003). Observations are presented every decade after 1950 and less frequently before that as a way of smoothing the series.

Here's where we see properties of growth rates in action:

The growth rate of GDP/person ...

is equal to the growth rate of GDP *minus* the growth rate of “person,” a.k.a. population

Figure 3.9: Population, GDP, and Per Capita GDP for the United States



Note: Data from Maddison (2003) and the Bureau of Economic Analysis. The average annual growth rate is reported next to each data series.